

Prepared By	09-Jun-2012	Mr.Chatchawan S.	 Renew coils Engineering & Supply Co., Ltd.	
Checked By	09-Jun-2012	Mr.Chatchawan S.		
Approved By	09-Jun-2012	Mr.Amnauy W.		
Certified By	09-Jun-2012	Mr.Amnauy W.		
Rev.	Date	Checked	Approved	Description
0	09-Jun-2012	Mr.Chatchawan S.	Mr.Amnauy W.	Issue for Approved

Renew Coils Engineering & Supply Co., Ltd.

Doc. No. : RNC-WPS-006-2012

Addendum

**WELDING PROCEDURE SPECIFICATION (WPS)
RNC-WPS-009**

WELDING PROCEDURE SPECIFICATION (WPS)

Company Name **Renew Coil Engineering & Supply Co., Ltd.**
WPS No. **RNC-WPS-009** Date **9-Jun-12** Revision No. **0**
Supporting PQR No.(s) **RNC-PQR-009** Date **9-Jun-12** Type(s) **Manual**
Welding Process(es) **Gas Tungsten Arc Welding + Shield Metal Arc welding** (Automatic, **Manual**, Machine, or Semi-Auto.)
(GTAW + SMAW)

JOINT (QW-402)

Joint Design **Single V**
Backing (Yes) **(No)** **X**
Backing Material (Type) **N/A**

☐ Metal ☐ Nonfusing Metal
☐ Nonmetallic ☐ Other

Notes _____

See Details Attachment sheet Joint Sketch

BASE METALS (QW-403)

P-No. **1** Group No. **1 or 2** to P-No. **1** Group No. **1 or 2**
Material Specification Type and Grade **A106 Gr.B, A234 WPBEquivalents** TO **A106 Gr.B, A234 WPBEquivalents**
Chem. Analysis and Mech. Prop. _____
Thickness Range :
Base Metal : Groove **1.6 mm -21.94 mm.** Fillet **All**
Pipe Dia. Range : Groove **NPS 2⁷/8"-Unlimited** Fillet **All**
Other _____

FILLER METALS (QW-404)

	GTAW	SMAW
Spec. No. (SFA)	5.18	5.1
AWS No. (Class)	ER 70S-G	E 7016
F-No.	6	4
A-No.	1	1
Size of Filler Metals	Ø 2.4 mm	Ø 3.2 & 4.0 mm.
Deposited Weld Metals	3.0 mm.	7.97 mm.
Thickness Range :		
Groove	6.0 mm Max.	15.94 mm. Max.
Fillet	ALL	ALL
Electrode-Flux (Class)	-	-
Flux Trade Name	-	-
Consumable Insert	-	-
Other	Kobelco TGS-50 or Equivalents	Kobelco LB 52 or Equivalents

POSITION (QW-405)

Position(s) of Groove **ALL**
Welding Progression **Up** **X** **Down**
Position(s) of Fillet **ALL**

POSTWELD HEAT TREATMENT (QW-407)

Temperature Range **595⁰-625⁰C**
Time Range **Holding Time 1 Hour**
Notes **Cooling Rate = 250⁰ Max.**

PREHEAT (QW-406)

Preheat Temp. Min. **T ≤ 25 mm. = 10⁰C**
Interpass Temp. Max. **T ≥ 25 mm. = 79⁰C**
Preheat Maintenance **250⁰C**
Notes **N/A**

GAS (QW-408)

	Percent Composition		
	Gas(es)	(Mixture)	Flow Rate
Shielding	Argon	99.99%	12-16 Ltr/Min
Trailing	-	-	-
Backing	-	-	-

WELDING PROCEDURE SPECIFICATION (WPS)

WPS No. RNC-WPS-009 Rev. 0

ELECTRICAL CHARACTERISTICS (QW-409)

Current ☒ AC ☒ DC Polarity EN(GTAW)+ EP(SMAW)
 Amps (Range) 100-140(GTAW) + 90-120(SMAW) Volts (Range) 10-13(GTAW)+ 27-37(SMAW)
 Tungsten Electrode Size and Type Ø 2.4 mm , Thoriated (EWTh-2)
 (Pure Tungsten, 2% Thoriated, etc.)
 Mode of Metal Transfer for GMAW N/A
 (Spray arc, short circuiting arc, etc.)
 Electrode Wire feed speed range N/A

TECHNIQUE (QW-410)

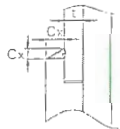
String or Weave Bead BOTH APPLICATIONS
 Orifice or Gas Cup Size 8.0 mm - 19.0 mm
 Initial and Interpass Cleaning (Brushing, Grinding, etc.) BRUSHING OR GRINDING
 Method of Back Gouging N/A
 Oscillation N/A
 Contact Tube to Work Distance N/A
 Multiple or Single Pass (per side) MULTIPASS
 Multiple or Single Electrodes SINGLE ELECTRODE
 Travel Speed (Range) 8-12 (GTAW)+5-10(SMAW)Cm/Minute
 Peening N/A
 Other N/A

Weld Layer(s)	Process	Filler Metal		Current		Volt Range	Travel Speed Range (Cm/min)	Other
		Class	Dia. (mm)	Type Polar.	Amp. Range			
1st	GTAW	ER 70S-G	2.4	DC EN	100-130	10-11	8-10	
2nd	GTAW	ER 70S-G	2.4	DC EN	120-140	10-13	10-12	
3 & Over	SMAW	E 7016	3.2 & 4.0	DC EP/AC	90-120	27-37	5-10	

We, the undersigned, certify that the statements in this record are correct and that the test welds were prepared, welded, and tested in accordance with the requirement of **ASME IX, 2007 latest Revision**.

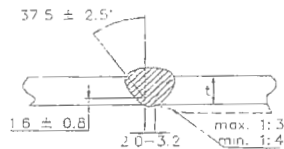
Prepared by: Mr. Chatchawan S. Signed: [Signature] Date: 9/06/12
 Reviewed by: Mr. Sansan Ch. Signed: [Signature] Date: 9-06-12
 Approved by: AMNUAY W. Signed: [Signature] Date: 29-06-12

ATTACHED FIGURE JOINT SKETCH



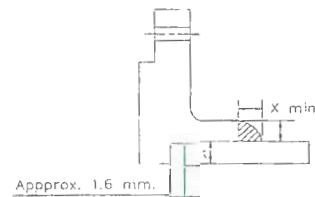
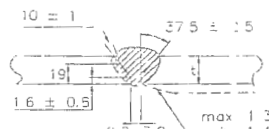
C_x (min.) = $1.25t$ but not less than 3.2 mm

JOINT No. A1



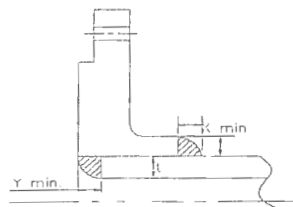
Misalignment (max.) = 1.5 mm

JOINT No. B1 & B2



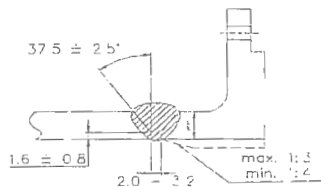
X min = The lesser of 1.4t or thickness of the hub.

JOINT No. A2



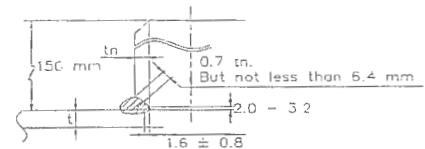
X min. = The lesser of 1.4t or thickness of the hub.
 Y min. = The lesser of t or 6.4 mm.

JOINT No. A3



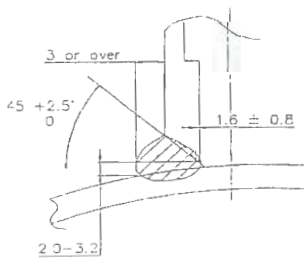
Misalignment (max.) = 1.5 mm.

JOINT No. B3

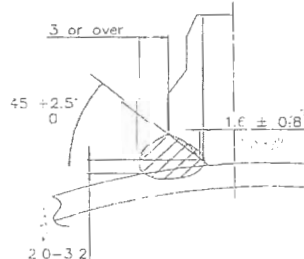


Misalignment (max.) = The lesser of 3.2 mm or 0.5tn.

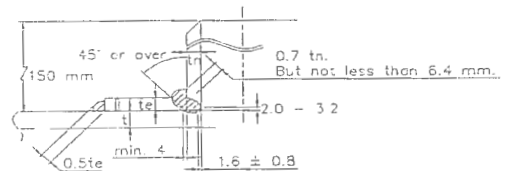
JOINT No. B4



JOINT No. C1

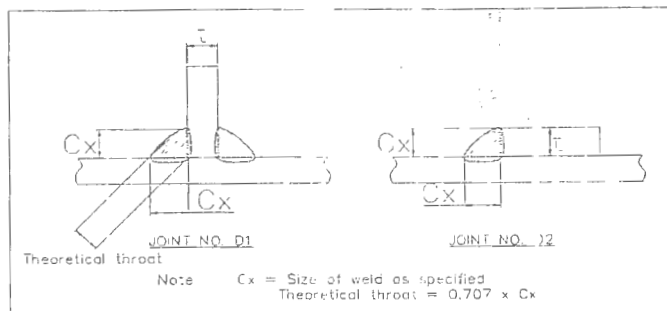


JOINT No. C2



Misalignment (max.) = The lesser of 3.2 mm or 0.5tn.

JOINT No. B5



Procedure Qualification Record (PQR)

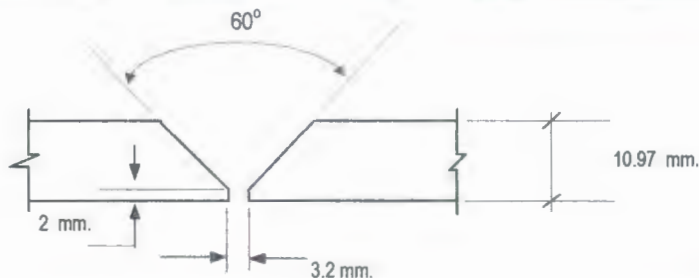
ASME-QW-200.2, Section IX, ASME Boiler Pressure Vessel Code

Record Actual Variables Used to Weld Test Coupon

RECORD OF WELDING (QW-483) PQR Number : **RNC-PQR-009**

Company Name :	RENEWCOILS ENGINEERING & SUPPLY CO.,LTD.		
PQR Record Number :	RNC-PQR-009	Date :	13 June 2012
WPS Number :	RNC-WPS-009		
Welding Process (es) :	Gas Tungsten Arc Welding (GTAW) + Shield Metal Arc Welding (SMAW)		
Type (Manual, Semi-auto, or Automatic) :	Manual		

JOINT DESIGN (QW-402)



(For combination qualifications, the deposited weld metal thickness shall be recorded for each filler metal or process used)

WELDING VARIABLES

BASE METALS (QW-403)				POSTWELD HEAT TREATMENT (QW-407)			
Material Spec.	A106	To	A106	Temperature	610 °C		
Type or Grade	Gr.B	To	Gr.B	Hold Time	1 HR.		
P. No.	P.1	To	P.1	Others			
Thickness of Test Coupon		10.97 mm.		GAS (QW-408)			
Diameter of Test Coupon		Ø 6"					
Others							
FILLER METALS (QW-404)				Percent Composition			
					Gas (es)	Mixture	Flow Rate
				Shielding	Argon	99.95%	15 L/Min.
				Trailing	-	-	-
				Backing	-	-	-
				ELECTRICAL CHARACTERISTICS (QW-409)			
SFA Specification		GTAW	SMAW	Current	DC (GTAW)	DC (SMAW)	
AWS Classification		ER70S-G	E7016	Polarity	EN	EP	
Filler Metal F-No.		6	4	Amperage	100-130 for GTAW and 110-130 for SMAW Voltage 10-11 for GTAW 27 for SMAW		
Weld Metal Analysis A-No.		1	1	Tungsten Electrode Size	Ø 2.4 mm. 2% thoriated.EWTh-2		
Size of Filler Metal		Ø 2.4 mm.	Ø 3.2 mm.	Others	N/A		
Others		Kobelco TGS-50	Kobelco LB 52	TECHNIQUE (QW-410)			
Weld Metal Thickness		-					
POSITION (QW-405)							
Position of Groove		6G		Travel Speed	8-11 cm/min.		
Progression of Welding (Uphill or Down)		Up Hill		String or Weave Bead	Both		
Others		N/A		Oscillation	N/A		
PREHEAT (QW-406)				Multipass or single Pass	Multiple Pass		
				Multipass or Single Electrodes	Multiple Pass		
				Others		N/A	
				Preheat Temperature		Ambient Temperature	
Interpass Temperature (max.)		219° C		RNC REPAIR QUALITY ENGINEERING QA/QC RNC			
Others		N/A					

Procedure Qualification Record

ASME-QW-200.2, Section IX (WPQR)

RECORD OF TEST RESULTS (QW-483) PQR Number:- **RNC-PQR-009**

Tensile Test Report No.TS-12-06-106

Specimen	Width (mm)	Thickness (mm)	Area (mm ²)	Ultimate Total Load (KN)	Ultimate Strength (N/mm ²)	Type of Failure & Loc.
2367/12TR1	19.63	10.42	204.545	106.264	519.519	OUT OF WELD
2367/12 TR2	19.39	10.67	206.891	105.284	508.886	OUT OF WELD

Guided Bend Tests (QW-160) Report No.BD-12-06-104

Type and Figure Number	Type of Bend	Indication	Results
2367/12 BS1	Transverse side bend	No Open Discontinuity	Accepted
2367/12 BS2	Transverse side bend	No Open Discontinuity	Accepted
2367/12 BS1	Transverse side bend	No Open Discontinuity	Accepted
2367/12 BS2	Transverse side bend	Open Discontinuity 0.7 mm	Accepted

Toughness Tests (QW-170)

Specimen Number	Notch Location	Notch Type	Test Temp.	Impact Values	Lateral Expansion		Drop Weight	
					% Shear	Mils	Break	No Break
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Fillet Weld Test (QW-180)

Result - acceptable : Yes N/A No N/A Penetration Into Base Metal : Yes N/A No N/A
Macroetch Test Results N/A

Other Tests

Type of Test Radiographic examination results Accepted Report No.PQR - 006
Hardness Testing Report No. HN-12-06-081
Macrostructure Report No. MA-12-06-103
Impact Testing Report No.IM-12-06-189

Welders Name Mr.Choochat M. Welder Number -
Tests Conducted By Mr.Keerati Sa-ngoen Laboratory Test LPN PLATE MILL

We certify that the statements in this record are correct and that the test coupons were prepared, welded, and tested in accordance with the requirements of section IX of the ASME Boiler and Pressure Vessel Code.

Manufacture or Contractor : 
Date : 12-06-12

Inspected by : Mr.Natthawut M.
Certified by : Mr.Chalermkiet R.
Date : 12-06-12 



PAE TECHNICAL SERVICE CO.,LTD
NDT, TESTING, INSPECTION & ENGINEERING CONSULTANT

69 SOI.OONNUT 64 (SUKSAMARN) SRINAKARIN RD, SUANLUANG,BANGKOK 10250

Tel : 02) 322-0222, Fax : 02)721-2577, RAYONG (038) 607-402-3

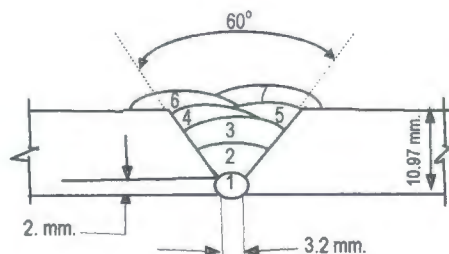
WELDER PROCEDURE QUALIFICATION RECORD

Page No. 1 of 1

Client : Renew Coils Engineering & Supply Co.,Ltd.	Project Name : PQR	Location : RNC Work Shop	Report No. PQR-006
	Customer Order No. : N/A	WPS No. : RNC-WPS-009	Date : 13-Jun-12

Welder No.	Welder Name	Welding Process	Test Position	Applicable Standard	Material		Electrode Specification	Results	
					Spec	Size		Visual	RT
-	MR.CHOOCHAT M.	GTAW+SMAW	6G	ASME IX	A106Gr.B	ø 6" Thk. 10.97 mm.	ER70S-G/E7016	ACCEPTED	ACCEPTED

Joint Details



Layer (S)	Process	Filler Metal		Current		Volt	Travel Speed cm/min	Interpas Temp °C	Heat Input (KJ / cm)	Shielding		Backing	
		Class	Dia. (mm.)	Type Polarity	Amp					GAS	Flow Rate (l/min)	GAS	Flow Rate (l/min)
1	GTAW	ER70S-G	2.4	DCEN	108	10	8	118	8.10	Argon	15	-	-
2	GTAW	ER70S-G	2.4	DCEN	138	11	11	121	8.28	Argon	15	-	-
3	SMAW	E7016	3.2	DCEP	117	27	8	194	23.69	-	-	-	-
4	SMAW	E7016	3.2	DCEP	119	27	10	205	19.27	-	-	-	-
5	SMAW	E7016	3.2	DCEP	115	27	8	211	23.28	-	-	-	-
6	SMAW	E7016	3.2	DCEP	113	27	8	214	22.88	-	-	-	-
7	SMAW	E7016	3.2	DCEP	114	27	8	219	23.06	-	-	-	-

PAE Inspection By

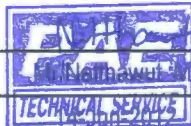
Witness By

Approved By

SIGNATURE :

NAME :

DATE :



SIGNATURE :

NAME :

DATE :



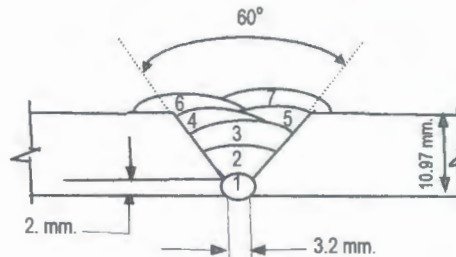
SIGNATURE :

NAME :

DATE :

Signature
ANUAY W.
29-06-12

Welding Sequences for RNC-PQR-009 (Butt Joint)



Weld Bead No.	Process	Class Filter Metal	Wire Size (mm)	Arc Current (Amps)	Arc Voltage (Volts)	Pot. AC/DC+	String or	Wire Feed	Rate of Travel (cm/min)	Heat Input (KJ/cm)	Gas Shielding Flow Rate (l/min)	Gas Backing Flow Rate (l/min)
							Weave Bead	Speed				
1	GTAW	ER70S-G	2.4	108	10	DCEN	-	-	8	8.10	15	-
2	GTAW	ER70S-G	2.4	138	11	DCEN	-	-	11	8.28	15	-
3	SMAW	E7016	3.2	117	27	DCEP	-	-	8	23.69	-	-
4	SMAW	E7016	3.2	119	27	DCEP	-	-	10	19.27	-	-
5	SMAW	E7016	3.2	115	27	DCEP	-	-	8	23.28	-	-
6	SMAW	E7016	3.2	113	27	DCEP	-	-	8	22.88	-	-
7	SMAW	E7016	3.2	114	27	DCEP	-	-	8	23.06	-	-



Renew Coils Engineering & Supply CO., LTD.

59 MOO 8 HIGHWAY 36, T. TABMA, A. MUANG, RAYONG 21000

TEL : (038) 663 - 476 FAX : (038) 633 - 477 MOBILE : (086) 340 -9978

Radiographic Examination Report

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Renew Coils Engineering & Supply CO., LTD.

59 MOO 8 HIGHWAY 36, T. TABMA, A. MUANG, RAYONG 21000

TEL : (038) 663 - 476 FAX : (038) 633 - 477 MOBILE : (086) 340 - 9978

Mechanical Test Report



Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415 , E-mail : qa@lpnmpm.co.th



21714

Tension Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.
Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Page 1 of 1
Report No. TS-12-06-106
Revision No. 0

Test Product	PQR No. RNC-PQR-009	Material Characterization	Pipe
Project Name	N/A	Test Method	ASTM A370-10 (Prepared by ASME IX : 2010)
Project No.	N/A	Standard of Test	ASME Sec. IX : 2010
Material Specification	A106 Gr.B	Type of Test	Reduced Section Tension
Dimension	Ø 6" Sch 80 (10.91 mm Thk.)	Equipment's Capacity	200 TONS
Date Received	19 June 2012	Equipment / Serial No.	SHIMADZU UH200A / 32098926
Date Tested	21 June 2012	Ambient Temperature	23 ± 5 °C

Item No.	Test No.	Specimen Dimension					Yield		Ultimate		Elongation (%)	Remark
		Diameter (mm)	Thickness (mm)	Width (mm)	Area (mm ²)	Gauge Length (mm)	Load (kN)	Strength (N/mm ²)	Load (kN)	Strength (N/mm ²)		
1	2367/12 TR1	N/A	10.42	19.63	204.545	N/A	N/A	N/A	106.264	519.519	N/A	OUT OF WELD
2	2367/12 TR2	N/A	10.67	19.39	206.891	N/A	N/A	N/A	105.284	508.886	N/A	OUT OF WELD

The unit of N/mm² is equivalent to MPa

Additional details : WPS No. RNC-WPS-009

Welding Process : GTAW+SMAW

Filler Metal/Electrode Spec't : ER-70S-G+E7016/TGS-50+LB52

Reported by :

(Mr.Keerati Sa-ngoen)

Laboratory Section Manager

Date 29 / 06 / 12



Approved by :

(Mr.Pavaret Preedawiphat)

Deputy Managing Director

Date 29 / 06 / 12

LPN PLATE MILL

- Notes**
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Testing Laboratory LPN Plate Mill Public Company Limited

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Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnmp.co.th



Bend Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Page 1 of 1

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Report No. BD-12-06-104

Revision No. 0

Test Product	PQR No. RNC-PQR-009	Material Characterization	Pipe
Project Name	N/A	Test Method	ASME Sec. IX : 2010
Project No.	N/A	Standard of Test	ASME Sec. IX : 2010
Material Specification	A106 Gr.B	Type of Jig	Guided-Bend Roller Jig
Dimension	Ø 6" Sch 80 (10.91 mm Thk.)	Diameter of Mandrel	40 mm
Date Received	19 June 2012	Bend Angle	180 Degree.
Date Tested	21 June 2012	Equipment / Serial No.	SHIMADZU UH200A/32098926

Item No.	Test No.	Type of Bend	Size of Specimen (ThickxWidthxLength)	Indication	Location of Indication	Remark
1	2367/12 BS1	Transverse side bend	10.00x10.61x161.96 mm	No Open Discontinuity	N/A	-
2	2367/12 BS2	Transverse side bend	10.00x10.61x161.48 mm	No Open Discontinuity	N/A	-
3	2367/12 BS3	Transverse side bend	10.08x10.61x161.28 mm	No Open Discontinuity	N/A	-
4	2367/12 BS4	Transverse side bend	10.08x10.77x161.43 mm	Open Discontinuity 0.7 mm	WELD ZONE	

Additional details : WPS No. RNC-WPS-009

Welding Process : GTAW+SMAW

Filler Metal/Electrode Spec't : ER-70S-G+E7016/TGS-50+LB52

Reported by : 

(Mr.Keerati Sa-ngoen)

Laboratory Section Manager

Date 29 / 06 / 12



Approved by : 

(Mr.Pavaret Preedawiphat)

Deputy Managing Director

Date 29 / 06 / 12

LPN PLATE MILL

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Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnplm.co.th



Impact Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Page 1 of 1

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Report No. IM-12-06-189

Revision No. 0

Test Product	PQR No. RNC-PQR-009	Material Characterization	Pipe
Project Name	N/A	Test Method	ASTM A370-10 and ASTM E23M-07a ¹
Project No.	N/A	Standard of Test	ASME Sec. IX : 2010
Material Specification	A106 Gr.B	Type of Test	Charpy V-Notch Impact
Dimension	Ø 6" Sch 80 (10.91 mm Thk.)	Equipment / Serial No.	Tinius Olsen / 221811
Date Received	19 June 2012	Equipment 's Capacity	542 Joules
Date Tested	22 June 2012	Ambient Temperature	23 ± 5 °C

Item No.	Test No.	Test Temperature (°C)	Size of Specimen			Results		Location of Test	Remark
			Thickness (mm)	Width (mm)	Length (mm)	Energy Absorbed (Joules)	Average (Joules)		
1	2367/12-1	-20°C	7.508	10.009	55.00	9.32	42.70	Base Metal	-
2	2367/12-2	-20°C	7.504	10.013	54.97	53.88			
3	2367/12-3	-20°C	7.503	10.005	55.00	64.89			
4	2367/12-1	-20°C	7.503	10.003	54.93	122.89	204.96	Weld Metal	-
5	2367/12-2	-20°C	7.500	10.000	54.99	237.12			
6	2367/12-3	-20°C	7.491	10.006	54.96	254.86			
7	2367/12-1	-20°C	7.502	10.005	54.90	122.89	138.69	HAZ	-
8	2367/12-2	-20°C	7.496	10.012	54.98	184.71			
9	2367/12-3	-20°C	7.501	10.007	54.92	108.48			

Additional details : WPS No. RNC-WPS-009

Welding Process : GTAW+SMAW

Filler Metal/Electrode Spec't : ER-70S-G+E7016/TGS-50+LB52

Reported by :

(Mr.Keerati Sa-ngoen)

Laboratory Section Manager

Date/...../.....

Approved by :

(Mr.Pavaret Freedawiphat)

Deputy Managing Director

Date/...../.....

Notes 1 The above results are valid exclusively for tested samples as mentioned in this report.

2. Partial publicity of the results on testing is prohibited without the written permission from LPN LABORATORY.

3. This report shall not be reproduced in whole or in part without the written approval of LPN LABORATORY.

LPN PLATE MILL



Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

Tel. 0 2815 6400-9 ext. 1179, 1210, Direct Line Fax. 0 2815 6415 , E-mail : qa@lpnpm.co.th

11066

Macrostructure Analysis Report

Page 1 of 1

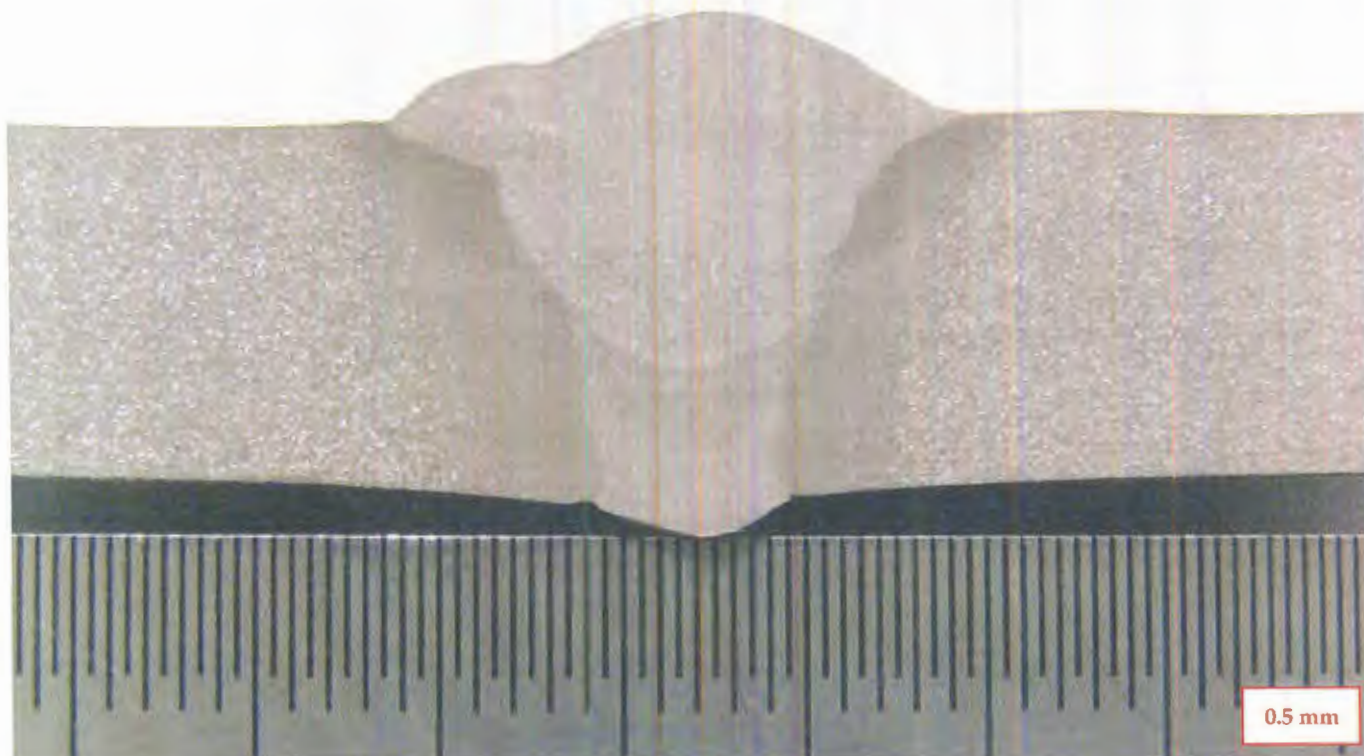
Customer : RENEWCOILS ENGINEERING & SUPPLY CO.,LTD.

Report No. MA-12-06-103

Address : 59 Moo 8 HIGHWAY 36 T.TABMA A.MUANG,RAYONG 21000 Thailand.

Revision No. 0

Test Product	PQR No. RNC-PQR-009	Material Characterization	Pipe
Test No.	2367/12 MA	Test Method	ASME IX : 2010
Project Name	N/A	Standard of Test	ASME IX : 2010
Project No.	N/A	Etchant Reagent	75ml H ₂ O+25ml HNO ₃ (65%)
Material Specification	A106Gr.B	Temperature	28 °C
Dimension	Ø6"Sch80(10.91 mm Thk.)	Time of Etching	5 minutes
Welding Process/Position	GTAW+SMAW	Surface Preparation	Polishing with sand paper
Type of Joint	Butt Joint		240, 400, 600, 800, 1000
Date Received	19 June 2012	Date Tested	21 June 2012



Additional details : The macrostructure shows the complete fusion.Weld metal and heat affected zone were free of cracks
Filler/Electrode Spec't : ER-70S-G+E7016 TGS-50+LB52

Reported by : *N. Sa-ngoen*

(Mr.Keerati Sa-ngoen)

Laboratory Section Manager

Date : 29/06/12



Approved by : *P. Preechawiphat*

(Mr.Pavaret Preechawiphat)

Deputy Managing Director

Date : 29/06/12

LPN PLATE MILL

Notes

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2. Partial publicity of the results on testing is prohibited without the written permission from LPN LABORATORY.
3. This report shall not be reproduced in whole or in part without the written approval of LPN LABORATORY.



Renew Coils Engineering & Supply CO., LTD.

59 MOO 8 HIGHWAY 36, T. TABMA, A. MUANG, RAYONG 21000

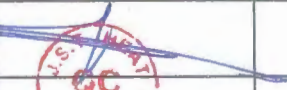
TEL : (038) 663 - 476 FAX : (038) 633 - 477 MOBILE : (086) 340 -9978

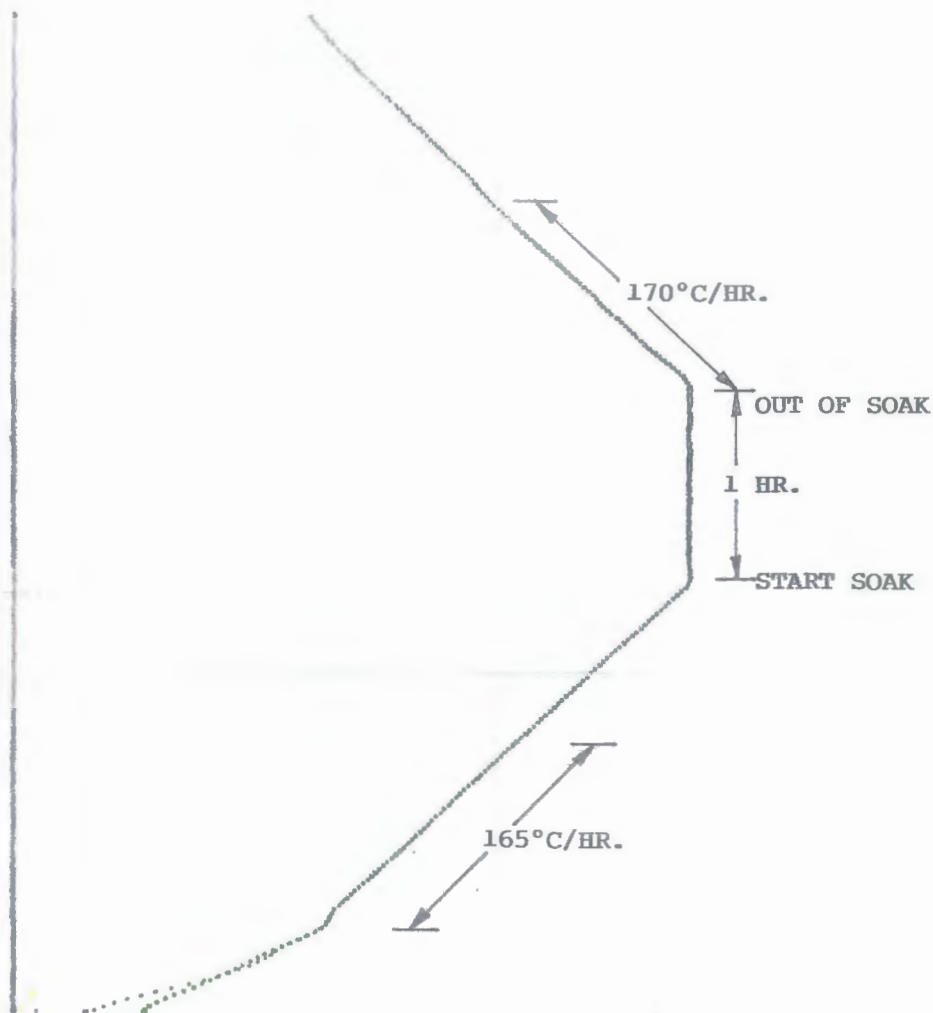
Post weld Heat Treatment Report

HEAT		J.S.T. HEAT TREATMENT COMPANY LIMITED	
POST WELD HEAT TREATMENT REPORT			
CLIENT : RENEW COILS ENGINEERING & SUPPLY CO. LTD.			
CONTRACTOR :	J.S.T. HEAT TREATMENT CO.,LTD.		PAGE : 1
PROJECT : PQR			
P.W.H.T.METHOD : ☒ LOCAL ☐ FURNACE			
COMPLETION DATE : 14-06-2012			REPORT NO.PW -001
CHART SPEED : ☒ 25 MM/Hr, ☐ 50 MM/Hr, ☐ 100 MM/Hr.			
RECORDER NO : EH-066A042			MATERIAL : A335 GR.6,A106 GR.B
LOCATION : ☒ SHOP JST. ☐ FIELD			UNIT :

HEAT TREATMENT CONDITION/RESULTS						
DESCRIPTION	LOADING TEMP(C°)	HEATING RATE (C°/Hr)	HOLDING TEMP (MIN-MAX)	HOLDING TIME (Hr)	COOLING RATE (C°/Hr)	UNLOADING TEMP (C°)
REQUEST	300	200	593-649	1 HR.	200	300
ACTUAL	300	165	610	1 HR.	170	300

DWG NO.,LINE NO.&CLASS,SHT.NO.SPOOL	WELD NO.	SIZE (")	THK. (MM)	WELD TYPE	T/C NO.
RNC-PQR-005		2"	3.91	BW	5
RNC-PQR-006		6"	7.11	BW	3,4
RNC-PQR-009		6"	10.97	BW	1,2
TOTAL	3	JOINTS	TOTAL	14	D.B.

CHECKED BY J.S.T. HEAT TREATMENT CO.,LTD.		REVIEWED BY	APPROVED BY	APPROVED BY
SIGN :				
NAME :	SUPACHAI K.	me Chaiwan		
DATE :	14-06-2012			



DATE. 14-06-2012

RECORDER NO. EH - 066A042

CHART NO. PW - 001

CHART SPEED. 25mm./HR.

CUSTOMER. RENEW COILS ENGINEERING & SUPPLY CO.,LTD.

PROJECT. PQR

DWG NO. RNC-PQR-005

SIZE. 2" THK. 3.91mm.

DWG NO. RNC-PQR-006

SIZE. 6" THK. 7.11mm.

DWG NO. RNC-PQR-009

SIZE. 6" THK. 10.97mm.

MATERIAL. A333 GR.6, A106 GR.B

START TIME. 21.30 PM

TECHNICIAN. MR. ANU S.

TC	COLOUR
1.	RED
2.	BLACK
3.	SKY BLUE
4.	GREEN
5.	BROWN



COOPERHEAT

Certificate of Calibration / Test



INNOVATORS IN THERMAL TECHNOLOGY

Certificate No.

JST-053/2012

Date of Issue

1/2/2012

Signatory (Print)

Supachai K

Equipment Description

RECORDER "CHINO" EH SERIES

Input or Sensor Type

"K"

Output or Range

0-1200°C

Cooperheat Identification

Serial No.

EH-066A042

Next Calibration Due

1/8/2012

Customer Name

Customer Order No.

Works Order No.

Test Parameter	200	400	600	800	1000	1200
Prior to Test	200	400	600	800	1000	1200
Post Test	200	400	600	800	1000	1200

Instrument Used	Model	Model	Date Calibrated
FLUKE	714	6986010	25 June 2011

All measurement are performed with equipment that is traceable to National Standard unless otherwise

Signature

HEAT

Date

1/2/2012

Customer

J.S.T. HEAT TREATMENT CO., LTD.

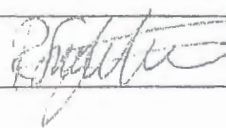
Certificate of Conformance

This material has been tested, inspected and approved by Cooperheat quality control in compliance with Cooperheat's BS EN ISO 9001 : 2000 Quality Management System.

43000 (100 mtrs)	TYPE "K" (NiCr-NiAl) Twin Thermocouple wire conforms to:- BS EN 60584-1. BS EN 60584-2: ANSI-MC96-1: Temperature Measurement Thermocouples. 1982
------------------	---

Batch No T1C7M18121142

QC Inspector's Sig:



Date

15/4/2008



CERTIFICATE OF CALIBRATION

mmc
(THAILAND) LTD

Issued By Marske Machine Hire & Calibration Centre

Date of Issue 25 June 2011

Certificate Number

E10544

Page 1 of 3 Pages



CALIBRATION
No. 0035

Marske Machine (Thailand) Co., Ltd
2nd Floor, The Seaboard Group Building
2/58 Highway 3191, MapTaPhut, Muang, Rayong 21150
Tel : (038) 682992-3, (038) 691563-4 Fax : (038) 682988
Email : sales@mmc-thai.com

Approved Signatory

A. M. P.
Marut L

Customer : J S T. Heat Treatment Co., Ltd
29/9 Soi Charoenmitr (Ekamai 10), Sukhumvit 63 Rd , Klongton Nue
Wattana, Bangkok 10110

Date Received 24 June 2011

Instrument -	System ID :	EC00745	Job Number :	0000113262
	Description :	Thermocouple Calibrator		
	Manufacturer :	FLUKE		
	Model Number :	714		
	Serial Number :	6986010	Last Certificate Number	E00544
	Procedure Version :	EP0008	Last Calibration Date	15/06/2010

Environmental Conditions

Temperature : 23°C +/- 2°C
Relative Humidity : 55% +/- 15%

Mains Voltage : 220V +/- 10V
Mains Frequency : 50Hz +/- 1Hz

Comments

The instrument was placed in lab and allowed to stabilise before calibrate
The direct comparison technique has been applied to calibrate.
All tests passed Mfg'Spec Model 714 Rev 1,10/97

Traceability Information

Instrument description	Serial number	Certificate number	Cal. Date	Cal. Period
3041 Precision Multi-Product Calibrator	F1031B8	19721	26/04/2011	52
3458A Multimeter	MY45043338	EE-0052-11	27/05/2011	52

Calibrated By : Tassanee K

Tassanee K

Date of Calibration : 25 June 2011

This certificate provides traceability of measurement to recognised National Standards and to the units of measurement realised at the National Physical Laboratory or other recognised National Standards laboratories

Copyright of this certificate is owned by the issuing laboratory and may not be reproduced except with the prior written approval of the issuing laboratory
This certificate complies with the requirements of BS EN ISO 10012 2003

Quality isn't Expensive It's Priceless

CERTIFICATE OF CALIBRATION



Certificate Number
E10544

Page 2 of 3 Pages

Test Title	Tolerance	STD Value	UUC Reading	% Of Spec.	Uncertainties
------------	-----------	-----------	-------------	------------	---------------

Calibration Information.

The uncertainty limits quoted refer to the measured values only, with no account being taken of the instrument ability to maintain its calibration.

A. M. P.

The reported expanded uncertainty of measurement is state as the standard uncertainty of measurement multiplied by a coverage factor $k=2$ corresponding to a coverage probability of approximately 95%.

Function : DC Voltage Measurement

Range 75 mVDC	19.8uV	10.000 0mV	10 000mV	0%	4uV
Range 75 mVDC	28.8uV	-10.0000mV	-9.990mV	35%	6uV
Range 75 mVDC	19.8uV	30.000 0mV	30 010mV	51%	4uV
Range 75 mVDC	19.8uV	50.000 0mV	50 010mV	51%	4uV
Range 75 mVDC	19.8uV	70.000 0mV	70.010mV	51%	4uV

Function : Temperature Measurement

1370 °C T/C Type K	0.83°C	-100.00°C	-99.7°C	36%	0.17°C
1370 °C T/C Type K	0.76°C	0.00°C	0.3°C	39%	0.16°C
1370 °C T/C Type K	0.74°C	100.00°C	100.2°C	27%	0.15°C
1370 °C T/C Type K	0.73°C	500.00°C	500.2°C	27%	0.15°C
1370 °C T/C Type K	0.76°C	1 000.00°C	1 000.3°C	39%	0.16°C
1200 °C T/C Type J	0.74°C	-100.00°C	-99.6°C	54%	0.15°C
1200 °C T/C Type J	0.70°C	0.00°C	0.3°C	43%	0.14°C
1200 °C T/C Type J	0.68°C	100.00°C	100.3°C	44%	0.14°C
1200 °C T/C Type J	0.68°C	500.00°C	500.3°C	44%	0.14°C
1200 °C T/C Type J	0.67°C	1 000.00°C	1 000.3°C	45%	0.14°C
1760 °C T/C Type R	2.5°C	0.00°C	1.0°C	40%	0.50°C
1760 °C T/C Type R	1.8°C	100.00°C	100.0°C	0%	0.36°C
1760 °C T/C Type R	1.4°C	500.00°C	500.0°C	0%	0.28°C
1760 °C T/C Type R	1.3°C	900.00°C	900.0°C	0%	0.26°C
1760 °C T/C Type R	1.2°C	1 300.00°C	1 300.0°C	0%	0.24°C

Function : DC Voltage Source

Range 75 mVDC	28.8uV	9.998mV	10.00mV	8%	6uV
Range 75 mVDC	28.8uV	29.998mV	30.00mV	8%	6uV
Range 75 mVDC	28.8uV	49.997mV	50.00mV	10%	6uV
Range 75 mVDC	28.8uV	69.997mV	70.00mV	10%	6uV

Function : Temperature Source

1370 °C T/C Type K	0.83°C	-100.27°C	-100.0°C	33%	0.17°C
1370 °C T/C Type K	0.76°C	-0.24°C	0.0°C	32%	0.16°C
1370 °C T/C Type K	0.74°C	99.86°C	100.0°C	19%	0.15°C
1370 °C T/C Type K	0.73°C	499.85°C	500.0°C	21%	0.15°C
1370 °C T/C Type K	0.76°C	999.88°C	1 000.0°C	16%	0.16°C
1200 °C T/C Type J	0.74°C	-99.98°C	-100.0°C	3%	0.15°C
1200 °C T/C Type J	0.70°C	0.07°C	0.0°C	10%	0.14°C
1200 °C T/C Type J	0.68°C	100.04°C	100.0°C	6%	0.14°C
1200 °C T/C Type J	0.68°C	500.04°C	500.0°C	6%	0.14°C
1200 °C T/C Type J	0.67°C	999.95°C	1 000.0°C	7%	0.14°C

CERTIFICATE OF CALIBRATION



Certificate Number
E10544

Page 3 of 3 Pages

Item Title	Tolerance	STD Value	UUC Reading	% Of Spec.	Uncertainty
0 °C T/C Type R	2.5°C	0.75°C	0.0°C	30%	0.50°C
0 °C T/C Type R	1.8°C	99.50°C	100.0°C	28%	0.36°C
0 °C T/C Type R	1.4°C	499.54°C	500.0°C	33%	0.28°C
0 °C T/C Type R	1.3°C	899.75°C	900.0°C	19%	0.26°C
0 °C T/C Type R	1.2°C	1 299.84°C	1 300.0°C	13%	0.24°C

of calibration.



Testing Laboratory LPN Plate Mill Public Company Limited

12408

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

Tel. 0 2815 6400-9 ext. 1179, 1210, Direct Line Fax. 0 2815 6415 , E-mail : qa@lpnpm.co.th

Hardness Testing Report

Page 1 of 1

Customer : RENEWCOILS ENGINEERING & SUPPLY CO.,LTD.

Report No. HN-12-06-081

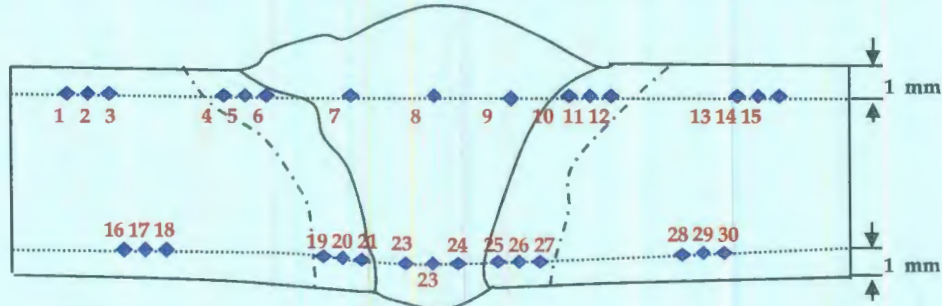
Address : 59 Moo 8 HIGHWAY 36 T.TABMA A.MUANG,RAYONG 21000 Thailand.

Revision No. 0

Test Product	PQR No. RNC-PQR-009	Material Characterization	Pipe
Test No.	2367/12 HN	Test Method	ASTM E92-82 : 2003
Project Name	N/A	Standard of Test	ASTM Sec. IX : 2010
Project No.	N/A	Type of test	Vicker Hardness Test
Material Specification	A106Gr.B	Equipment	Mitutoyo, HV-115
Dimension	Ø6"Sch80(10.91 mm Thk.)	Indenter Size	Diamond, pyramid 136 deg.
Welding Process/Position	GTAW+SMAW	Test Load	HV 10 Kgf.
Type of Joint	Butt Joint	Total Force Dwell Time	15 s
Date Received	19 June 2012	Cal. Block	No. 62137 No. A62072
Date Tested	21 June 2012	Ambient Temperature	25.8 °C

Test Location

Total = 30 points



Point No.	Location	Hardness value Value (Scale HV)	Point No.	Location	Hardness value Value (Scale HV)	Point No.	Location	Hardness value Value (Scale)
1	BASE	150.8	16	BASE	143.5	*****		
2	BASE	152.7	17	BASE	142.1			
3	BASE	152.4	18	BASE	144.0			
4	HAZ	156.4	19	HAZ	159.5			
5	HAZ	161.1	20	HAZ	160.2			
6	HAZ	168.6	21	HAZ	159.9			
7	WELD	167.4	22	WELD	142.4			
8	WELD	153.4	23	WELD	142.8			
9	WELD	194.5	24	WELD	142.7			
10	HAZ	177.4	25	HAZ	162.9			
11	HAZ	164.0	26	HAZ	166.0			
12	HAZ	154.6	27	HAZ	162.4			
13	BASE	154.9	28	BASE	151.9			
14	BASE	154.1	29	BASE	152.2			
15	BASE	153.7	30	BASE	151.4			

Additional details : Filler/Electrode Spec't : ER-70S-G+E7016 TGS-50+LB52

Reported by :
(Mr.Keerati Sa-ngoen)
Laboratory Section Manager
Date : 29/06/12



Approved by :
(Mr.Pavard Preedawiphat)
Deputy Managing Director
Date : 29/06/12

Notes

1. The above results are valid exclusively for tested samples as mentioned in this report.
2. Partial publicity of the results on testing is prohibited without the written permission from LPN LABORATORY.
3. This report shall not be reproduced in whole or in part without the written approval of LPN LABORATORY.

LPN PLATE MILL



Renew Coils Engineering & Supply CO., LTD.

59 MOO 8 HIGHWAY 36, T. TABMA, A. MUANG, RAYONG 21000

TEL : (038) 663 - 476 FAX : (038) 633 - 477 MOBILE : (086) 340 - 9978

Consumable Certificate

KOBELCO

24/10/54

INSPECTION CERTIFICATE

PURCHASER

Date of Issue 11/Aug/2011

Certificate No. TF11-035

Date of Inspection 16/Aug/2011

TRADE DESIGNATION	SIZE (mm)	MFG. No.	APPLICABLE SPECIFICATION & CLASSIFICATION
LB-52	3.2 x 350	12110	AWS A5.1 E7016

Chemical Composition (%)

Elements	C	Si	Mn	P	S	Ni	Cr	Mo	V	*1*
Weld Metal	0.07	0.52	0.93	0.009	0.003	< 0.01	0.03	< 0.01	0.009	0.97

Tensile Test of Weld Metal				Impact Test of Weld Metal			
Yield Strength at 0.2% offset	Tensile Strength	Elongation	Test Temp.	Absorbed Energy		Tested size for Mechanical test: 4.0x400 mm Tested MFG. No. of Mechanical test: 12009	
465 MPa	554 MPa	33 %	-30 °C	Avg.	246 J		

Welding Conditions				Postweld Heat Treatment	REMARKS
Type of Current	AC			--- °C x --- Hr.	*1*: Mn+Ni+Cr+Mo+V
Welding Current	120 A				
Welding Voltage	24 V				

We hereby certify that the test results of the above welding material are as described herein and satisfy the requirements of applicable specification.



THAI-KOBE WELDING CO., LTD.

CHIEF INSPECTOR

[Signature]



KOBELCO

INSPECTION CERTIFICATE

CHASER

Date of Issue

17/02/2011

Certificate No.

KF11-002

Date of Inspection

15/Feb/2011

TRADE DESIGNATION	SIZE (mm)	MFG. No.	APPLICABLE SPECIFICATION & CLASSIFICATION
TC-S50	2.4	FOB2704	AWS A5.18 ER70S-C

Chemical Composition (%)

Element	C	Si	Mn	P	S	Cu	Ni	Cr	Mb	V
WIRE	0.10	0.75	1.39	0.011	0.017	0.13	0.01	0.01	< 0.01	0.002

Tensile Test of Weld Metal

Impact Test of Weld Metal

Yield Strength at 0.2% offset	Tensile Strength	Elongation	Test Temp.	Absorbed Energy	Tested size for Mechanical test: 2.4 mm
425 MPa	545 MPa	38 %	-30 °C	Avg. 137 J	Tested SIFA No. of Mechanical test: P9B4-140

Working Conditions

REMARKS

Welding Process	DCEN	Shielding Gas	AT
Welding Current	120 A		
Welding Voltage	14 V		

We hereby certify that the test results of the above welding material are as described herein and satisfy the requirements of applicable specification.



KOBE MIG WIRE (THAILAND) CO., LTD.



CHIEF INSPECTOR

[Signature]

10001



Renew Coils Engineering & Supply CO., LTD.

59 MOO 8 HIGHWAY 36, T. TABMA, A. MUANG, RAYONG 21000

TEL : (038) 663 - 476 FAX : (038) 633 - 477 MOBILE : (086) 340 - 9978

Material Certificate

ArcelorMittal South Africa Limited
Tubular Products
Genl. Hertzog rd.
P O Box 48 Vereeniging 1930
South Africa

MATERIAL TEST CERTIFICATE

SEAMLESS TUBE

Telephone +27 (0)16 450 4220
Fax +27 (0)16 423 4906



Under
API Spec 5L and SCT
SL - 0319
SCT - 0419



ISO 9001: 2008
DE-395997 QM

Test Certificate
EN 10204:2004 TYPE 3.1

ArcelorMittal

Customer: **UR MENA LIMITED**
Order No: **4000013258**
Certificate Reference No: **040060442738**
Product: **FULLY KILLED HOT FINISHED CARBON STEEL SEAMLESS TUBES**
Specification: **ISO3183:2007/API 5L:2007 L245/L290/B/X42 PSL1 ASTM A106.08B/A53B.07/ A530.04 ASME SA106.08B/SA53B.07/SA530.04**
Product Marking: **ARCELORMITTAL SA ISO3183/Spec 5L-0319 MONOGRAM 09-11 ASTM/ASME A/SA106B A/SA53B 168.300 (6.626) 7.110 (0.280) 6.000 (19.700) L245/B L290/X42 PSL1 SMLS TESTED 20.7 MPa (3000 psi) CAST NO:99B3586 PROD/O NO: T7941325810**

Customer Order/Contract No: **STH9 137038**
Material No: **1000000467**
Cast/Heat No: **99B3586**

Page 1 of 1

General Information

Quantity	Mass	Dimensions			Total Length
		Tube OD	Thickness	Length	
38	6,443.660(kg)	168.300(mm)	10.970(mm)	6.000(m)	228.000(m)
	14,205.693(lb)	6.626(")	0.431(")	19.700 (ft)	748.600 (ft)

Steel making process	Final Rolling Operation
Electric Arc	Final hot rolling operation finished above 860°C and cooled in still air.

Chemical Composition

Chemical Composition			R22-(V + Nb + Ti)					R24-(Nb + V)																
Element(%)	C	Si	Mn	S	P	Cr	Ni	Mo	Cu	V	Al	Ti	Sn	Ca	N	B	Nb	CE	R22	R24				
Minimum	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-				
Maximum	0.200	-	1.30	0.030	0.030	0.50	0.50	0.150	0.500	-	-	-	-	-	-	-	-	0.41	0.15	0.060				
Heat	0.158	0.31	0.80	0.001	0.004	0.07	0.09	0.025	0.090	0.001	0.050	0.001	0.007	0.0017	0.0047	0.0002	0.0010	0.32	0.00	0.002				
Product	0.158	0.310	0.800	0.001	0.004	0.070	0.090	0.025	0.090	0.001	0.050	0.001	0.007	0.002	0.005	0.000	0.001	0.323	0.003	0.002				
Product (ADD)	0.158	0.310	0.800	0.001	0.004	0.070	0.090	0.025	0.090	0.001	0.050	0.001	0.007	0.002	0.005	0.000	0.001	0.323	0.003	0.002				

Mechanical Properties

Specification	UTS (Rm)		Yield (0.5%)		% EL
	MPa	psi	MPa	psi	
Limits	415	60000	290	42000	30.0
Minimum	-	-	-	-	-
Maximum	-	-	-	-	-
(1) Actual	485	70343	324	46992	34.0
(2) Actual	481	69763	321	46557	35.0
(3) Actual					
(4) Actual					

	UTS (Rm)		Yield (0.5%)		% EL
	MPa	psi	MPa	psi	
(5) Actual					
(6) Actual					
(7) Actual					
Orientation & type of tensile test			Longitudinal, SAMP		
Width of tensile piece (mm)			25.4mm		
Orientation of impact test piece					

OTHER TESTS	
Flattening	Passed
Hydrostatic	20700 kPa (3000 psi) - 5 sec
NDI: EMI	PASS - ASTM E570 - 12.5% NOTCH
NDI: UT	UT not required
HV 10 Kg	144 147 157

Remarks:

Material in accordance with NACE MR0175:2003/ISO15156-2:2003, MR0103:2010. Dimensions to ANSI B36.10M-2004
The material conform to the hot yield strength requirements as per ASME, Sect II, Pt D, Table Y-1, 2010
All the material conform to the visual and dimensional requirements and is made to a suitable fine grain practice

Quality Assurance Manager: **R. Bester**

Date of Issue: **2011.10.03**

BY **UR MENA LIMITED**

PURCHASE ORDER NO. **45014262**

ITEM NO. **1**

Certified by: **R**

We hereby certify that the steel grade and quality level of all products are in conformity with the order and comply fully with specification requirements. No changes, amendments or additions may be made to this document. Any changes which are effected shall invalidate this certificate.